FEATURE

Designing Water Utility Cooperative Agreements: Lessons Learned in North Carolina

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Key Takeaways

Cooperation among water utilities expands opportunities to provide an affordable, reliable water supply, both now and in the long term.

Computational modeling of utility partnerships can explore the potential benefits (and pitfalls) of cooperative infrastructure development.

Analysis of agreements for shared treatment plant development in North Carolina provides a framework for design and evaluation of cooperation in other utility contexts.

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Developing infrastructure projects for water and wastewater utilities can take years or even decades. After environmental reviews, master plan updates, and suitability analyses are complete, short-listed options may still face hurdles in construction, financing, implementation, and operation. As urban areas continue to grow and opportunities for freshwater supply expansion shrink, the challenges of keeping up with rising demand increase. Under these circumstances, how can water managers ensure affordable, reliable water service over the long term?

Growth and service area expansion also mean that municipalities and utilities become pressed to share water resources. Successful resource management among multiple water systems can require substantial political, legal, and institutional cooperation, adding to already complicated long-term planning for each partnering utility. However, within the constraints of cooperation come practical benefits. Cooperative infrastructure development can reduce capital and operating costs through economies of scale, make more efficient use of existing resources, and build physical redundancies for emergency conditions (i.e., additional water treatment or supply sources). As a result, interutility agreements are becoming more common in the United States and worldwide.

Given the unique context that each water utility faces, the details of cooperative agreements inevitably vary.

Within cooperative partnerships, it is also vital to understand the stability of each partner utility and explore the subtleties that could make or break an agreement.

This makes agreements difficult to generalize and highlights the need for research into their effectiveness under uncertainty. To that end, our team at the University of North Carolina at Chapel Hill and Cornell University (Ithaca, N.Y.) have collaborated with water utilities of the Research Triangle region of North Carolina to characterize the benefits and drawbacks of interlocal cooperation.

In particular, four of the region's utilities, serving hundreds of thousands of residents, plan to jointly construct, finance, and operate a shared water treatment plant. The following sections explore our findings on cooperative infrastructure financing for water utilities, work initially published in *Journal AWWA* in 2019 (Gorelick et al. 2019) and followed up by a study published in *Water Resources Research* (Gorelick et al. 2022), which offer lessons learned and reveal sometimes surprising outcomes arising from joint long-term development.

North Carolina's Research Triangle: Four Utilities, One Treatment Plant

The Research Triangle is home to more than two million residents, anchored by the three population centers of Raleigh, Durham, and Chapel Hill and expanding into three surrounding communities of Cary, northern Chatham County, and Pittsboro (Figure 1). Chapel Hill is served by the Orange Water & Sewer Authority (OWASA).

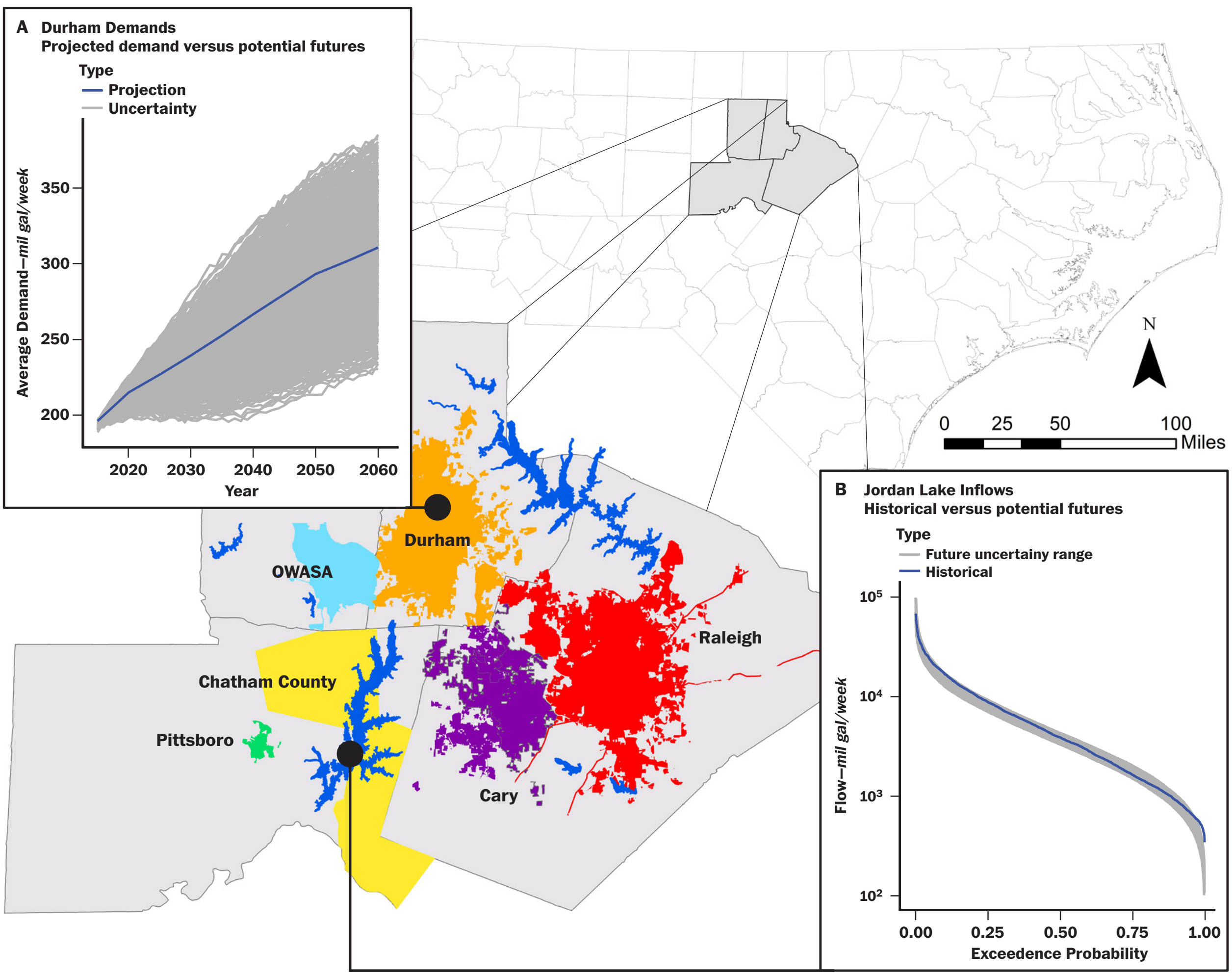
By 2060, regional water utilities for these communities expect demands to rise considerably. For example, the blue line in inset A of Figure 1 shows how Durham's demands are projected to grow from about 200 mil gal/week in 2015 to more than 300 mil gal/week by 2060, greater than 50% growth in 45 years.

To meet demands, each Triangle utility maintains a portfolio of primarily surface water supplies. Of the region's supply sources, Jordan Lake (labeled in inset B of Figure 1) is the largest. Five of the six Triangle utilities, Raleigh excluded, hold surface water allocations in Jordan Lake, making its sustainability paramount to regional supply reliability. Despite major droughts in 2002 and 2008, Jordan Lake has remained a reliable source of drinking water for Triangle consumers. Though Cary, Chatham County, Durham, and OWASA hold allocations to Jordan Lake storage, only Cary has an intake to directly access the lake's supply.

Over the next half-century, however, water supply and demand conditions are expected to deviate from historical trends. Climate change and land use urbanization in the Triangle call into question the future availability and predictability of surface water supply for reservoirs. Population growth, even with reductions in per-capita water use, will likely bring rising demands. In contrast, the potential for economic recession or water reuse innovation could lead to significant drops in demand over the longer term. Gray bands in inset plots of Figure 1 demonstrate how these wide-ranging uncertainties in water supply and demand could unfold over the planning period for Triangle utilities (today to 2060).

To ensure that water supply remains reliable and affordable in the future, utilities in the Triangle (and globally) will need to rely on both demand

Geography of North Carolina Research Triangle Regional Utility Service Areas



OWASA—Orange Water & Sewer Authority

Colors show service areas for Cary, Chatham County, Durham, Orange Water & Sewer Authority, Pittsboro, and Raleigh. Insets show examples of uncertainty ranges in future water demands for Durham (inset A) and future inflows to Jordan Lake between historical conditions and 2060 (inset B).

Figure 1

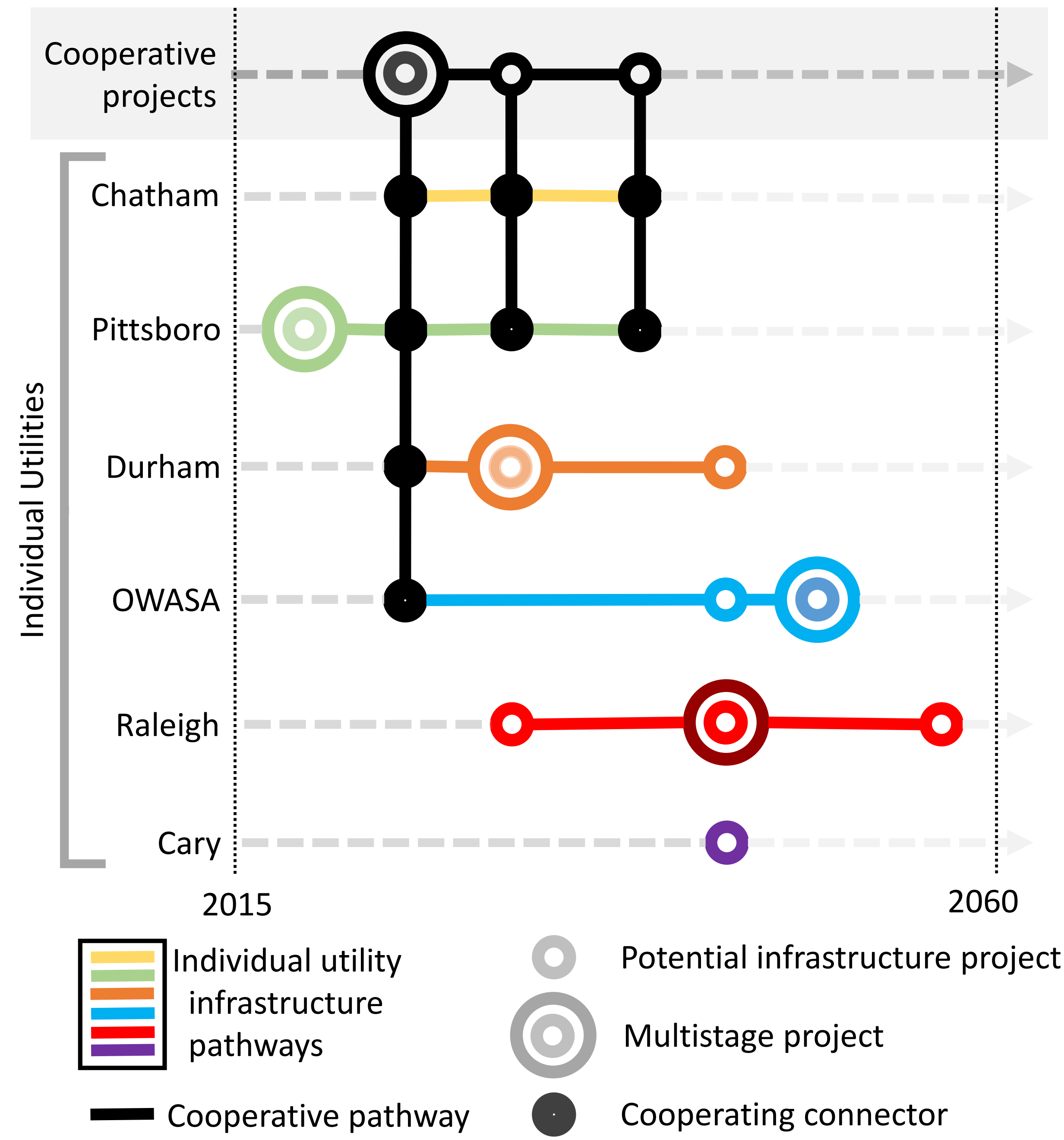
management and expansion of supply infrastructure. Four of the Triangle's utilities—Chatham County, Durham, OWASA, and Pittsboro—are planning to leverage their underutilized Jordan Lake allocations to boost long-term supply reliability. Doing so requires new treatment and transmission infrastructure for each community. Rather than building independent facilities, the four partner utilities formed the Western Intake Partnership (WIP) to explore the development of a shared water treatment plant on Jordan Lake.

Many Planning Options, Many Uncertainties

Electing to cooperate might seem an easy and prudent planning choice, but implementation, operation, and financing decisions for the WIP will be complex. Several questions have loomed large, including these:

- How much treatment capacity does each partner need to meet demands in 2030, and will their respective demands grow asymmetrically over the planning horizon (to 2060)?

Potential Water Supply Infrastructure Development Pathways for Research Triangle (N.C.) Water Utilities



OWASA—Orange Water & Sewer Authority

Schematic includes regional cooperative pathways (top row) over the planning horizon. Vertical (black) lines indicate potential project connections between cooperating utilities. Circles indicate estimated year in which projects become available to implement (e.g., after clearing environmental permitting processes, board approval, and so on).

Figure 2

- How should different initial claims on demand and different growth rates translate into costs?
- Can use agreements be adjusted over time to account for unforeseen conditions (e.g., extreme drought or demand growth)?

Though the WIP collaboration is a significant effort, the shared water treatment plant (WTP) it hopes to develop on Jordan Lake will be just one project in a pathway of capital improvement projects for each utility. Figure 2 lays out the potential infrastructure expansion projects planned for all six Triangle utilities before 2060.

In Figure 2, the WIP Jordan Lake WTP is represented by the concentric circles in black in the top row—the project is currently being considered in multiple stages, like a number of other Triangle infrastructure projects, which could be expanded in the future after initial construction. While most utilities in the Triangle have additional supply infrastructure options to consider independently of other utilities, smaller communities like Chatham County and Pittsboro plan to primarily rely on cooperative projects to meet long-term demands (marked by vertical black lines in Figure 2).

As detailed by Gorelick and colleagues (2022), it is difficult to identify a single "optimal" set of future decisions that a utility may make—there are just too many uncertainties. Instead, it can be informative for water managers to focus on what decisions or rules that trigger actions perform best under uncertainty; for example, when reservoir levels drop below a certain target, a utility implements its drought response plan. Identifying these more "robust" actions can simplify decision-making by isolating what choices are least likely to contribute to unacceptable water supply or financial stress.

Quantifying Regional Cooperation: How, Why, and What's Next

Assessing the effects of many layered and interdependent uncertainties requires a computational experiment of scale to match. To identify effective, robust regional strategies for water utilities in the Triangle, millions of future simulations were run to evaluate how cooperative actions influence the supply reliabilities and financial risks for all regional utilities.

"Bottom-Up" Modeling Approach to Reveal Robust Decision-Making Strategies

By capturing a wide range of future conditions—and limiting assumptions that the likelihood or probability of specific scenarios are well known—we explored not only what actions best insulate utilities from risk but also what combinations of factors (e.g., high demands combined with low water levels) tend to stress utility performance and cooperative agreements the most. Figure 3 summarizes the key aspects of this research.

Because it is important to weigh supply reliability against financial risk (and other interests) when setting utility management policy, a multi-objective search was used to identify a large and diverse set of potential decision-making strategies. These strategies capture the "optimal tradeoffs" across six objectives; improving a given strategy's performance in one objective often degrades performance in one or more of the other five objectives—in other words, there are explicit tradeoffs. In part A of Figure 3, each gray dot is one of these potential strategies, and the performance shown is based on two key objectives of supply reliability and the frequency of water use restrictions.

> **Whether seeking to meet supply or financial challenges, interutility agreements offer a lifeline in long-term planning where other options may be out of reach.**

To Meet Utility Performance Goals, Cooperation Is Key

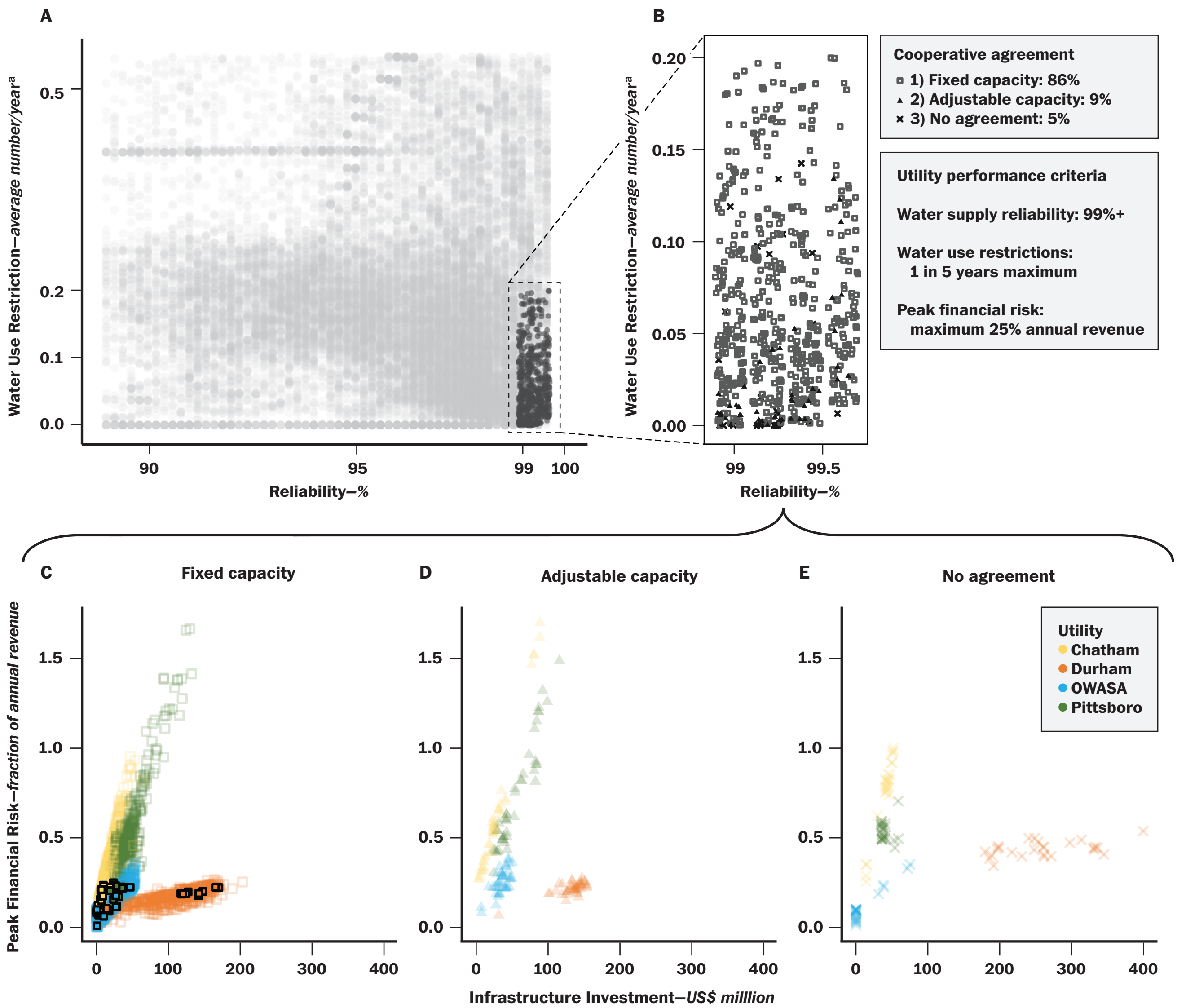
Having thousands of potential strategies is exciting but also overwhelming. However, this larger view of what is possible in terms of tradeoffs in key performance objectives narrows with consideration of what outcomes regional utilities prefer or are willing to accept. Preferences quickly filter the analysis down to a smaller subset of results that are relevant to decision makers. For water utilities in the Triangle, potential strategies were screened using the following criteria:

- Maintain very high reliability—99% of years without a storage failure
- Low use-restriction frequency—no more than once in five years on average
- Low peak financial risk—unexpected short-term expenditures being no more than 25% of annual revenues in any given year

These strategies are shown in black in Figure 3, part A, and part B zooms in to better understand their context.

From thousands of potential management strategies, only a few hundred were able to satisfy criteria of supply reliability and use restriction (Figure 3, part B). Of those, 95% of the satisfactory strategies took advantage of cooperative agreements through the WIP, shown as squares and triangles in Figure 3, part B. It should be noted that a handful of strategies met utility performance criteria without cooperative agreements (marked by "x" icons in Figure 3, part B), but these relied heavily on alternative infrastructure projects and resulted in significantly higher costs.

Study Results and Screening Techniques to Identify Effective Management Strategies



OWASA—Orange Water & Sewer Authority

[a]For example, one in five years would have a fraction of 0.2, and a fraction of 0.5 indicates restrictions happening every other year on average.

Part A shows all identified solutions, with those meeting utility reliability and conservation performance criteria in black; part B, type of cooperative agreements (shapes) available for solutions from part A that meet performance criteria; parts C–E, future infrastructure investment (*x*-axis) and peak annual financial risk (*y*-axis) of solutions in part B, separated by agreement type (by panel) and by utility (by color). In part C, solutions with black outlines indicate those meeting the financial risk performance criterion—no solutions in parts D or E satisfy this criterion for all utilities simultaneously.

Figure 3

According to these findings, regional and individual cooperation are essential to meet utility supply and financial goals.

Less Complex, Fixed Agreements Tend to Win Out, Providing Financial Stability

A complication of WIP planning, and likely of any cooperative utility agreement, is the asymmetrical size and growth potential of each partner. These differences are visible in the distributions of financial outcomes under different strategies for each utility—shown in color in parts C, D, and E of Figure 3. As the largest population center among WIP partners, Durham most often experiences the largest infrastructure investment costs.

High costs for Durham are exacerbated when no cooperative agreement is reached (Figure 3, part E), meaning Durham is essential to the WIP succeeding and incentivized to participate. Pittsboro and Chatham County, alternatively, see the largest variability in the peak financial risk they may experience, despite lower total infrastructure investment than Durham. This is because both are currently small and projected to grow rapidly; under futures where demands do not grow as projected, an unexpectedly small customer base makes it difficult for Chatham County or Pittsboro to raise revenues that match their debt obligations, giving rise to significant financial risk. Together, these divergent outcomes speak to a need to consider tradeoffs among agreement partners, no matter the agreement structure involved.

Within cooperative partnerships, it is also vital to understand the stability of each partner utility and explore the subtleties that could make or break an agreement. One such aspect is the structure of the cooperative contract itself. In the case of the WIP, partners can choose to enter financial agreements to construct and operate a shared Jordan Lake WTP in multiple ways.

On the basis of the broad types of interutility agreements in place across the United States, as explored by Gorelick and colleagues (2019), cooperative development was modeled here following either (1) a fixed-capacity or (2) an adjustable-capacity agreement. For the WIP, that means each utility would receive a(n) fixed (adjustable) allocation of treatment capacity in a shared Jordan Lake WTP and pay a(n) fixed (adjustable) proportion of the project's capital financing.

In parts C, D, and E of Figure 3, strategies using these WIP agreements are compared with those without WIP cooperation in terms of total infrastructure investment and peak financial risk for each utility of the WIP. Somewhat surprisingly, focusing on contract types revealed that not only do the vast majority of strategies that meet utility reliability and use-restriction goals rely upon a fixed capacity agreement for the WIP, but a fixed agreement is the only cooperative option that also meets performance criteria of low peak financial risk (points in Figure 3, part C, with bold black outlines).

With an adjustable agreement, a utility can update its capacity allocation and proportional debt service obligations, should demands shift away from initial projections. Rather than reducing financial risk, however, this mutual flexibility exposes WIP partners to the risks of their partners' demand projections also being inaccurate. As a result, interutility cooperation with fixed allocations allowed for more stable water supply and financial planning.

Gaining From Cooperation

Water and wastewater utilities can gain a great deal from cooperation. Whether seeking to meet supply or financial challenges, interutility agreements offer a lifeline in long-term planning where other options may be out of reach. More and more in the United States, utilities are relying on cooperative agreements, making it crucial to ensure they are robust in the face of potential hydrologic, demand, and institutional uncertainties. This article summarizes years of research on this topic, with the objective of providing insight on improving cooperative development that may benefit water managers, utility planning staff, and policymakers.

Acknowledgment and Data Availability

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